

Supplementary Materials: A Gauge-Free Predictive Model for the Tidal-Sampling Bias of Sun-Synchronous Satellites, Validated on Macrotidal Tidal Flats

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This document accompanies the main manuscript and contains all supplementary figures, the extended Table S1, and pointers to the public derived-data tables that support the reported statistics. Section, equation and figure numbers in the main manuscript are referenced by their numerical labels without an “S” prefix; entries in this document use the “S” prefix.

Table S1 — Stability regression coefficients

Regression coefficients of the closed-form model $\text{mean bias} = \beta \cdot \mathbf{A} \cdot \langle \cos \theta \rangle$ (Equation 1 of the main text) when the 15 (site $\times$ sensor) sample is repartitioned by acquisition year, season, sensor, or via leave-one-site-out cross-validation. n is the number of (site $\times$ sensor) points contributing to each fit. The pooled bootstrap mean / 95 % CI is reported on the final row.

Table S1a — Annual partition

		Intercept						RMSE	MAE
Year	n	(m)	Slope β	r	R ²	p-value	SE(β)	(m)	(m)
2020	9	−0.154	1.625	0.920	0.846	0.0004	0.262	0.177	0.139
2021	9	−0.165	1.718	0.939	0.882	0.0002	0.238	0.118	0.100
2022	13	−0.173	1.517	0.948	0.900	<0.0001	0.153	0.126	0.105
2023	13	−0.200	1.316	0.974	0.949	<0.0001	0.092	0.103	0.084
2024	13	−0.203	1.345	0.920	0.847	<0.0001	0.173	0.131	0.102

¹⁹ **Table S1b — Seasonal partition (meteorological seasons, KST)**

		Intercept						RMSE	MAE
Season	n	(m)	Slope β	r	R ²	p-value	SE(β)	(m)	(m)
DJF	14	−0.116	1.491	0.971	0.943	<0.0001	0.106	0.180	0.141
MAM	12	−0.084	1.138	0.992	0.984	<0.0001	0.046	0.077	0.068
JJA	12	−0.269	0.809	0.919	0.845	<0.0001	0.110	0.118	0.094
SON	13	−0.555	1.163	0.924	0.854	<0.0001	0.145	0.245	0.188

²⁰ **Table S1c — Per-sensor fit**

		Intercept						RMSE	MAE
Sensor	n	(m)	Slope β	r	R ²	p-value	SE(β)	(m)	(m)
Landsat	5	−0.044	1.734	0.993	0.986	0.0007	0.121	0.065	0.057
8									
Landsat	5	−0.063	2.007	0.998	0.996	0.0001	0.072	0.031	0.028
9									
Sentinel-	5	−0.048	1.712	0.999	0.997	<0.0001	0.053	0.025	0.020
2									

21 **Table S1d — Pooled bootstrap ($n = 15$; 2 000 resamples, seed = 42)**

Parameter	Mean	Median	95 % CI low	95 % CI high
Slope β	1.764	1.777	1.443	1.912
Intercept (m)	−0.070	−0.059	−0.212	−0.029

22 **Table S1e — Leave-one-site-out cross-validation (per-point residuals)**

Held-out site	Sensor	$A \cdot \langle \cos \theta \rangle$ (m)	Measured bias (m)	Predicted bias (m)	Residual (m)
Ganghwa-do	L8	−0.616	−1.133	−1.166	+0.033
Ganghwa-do	L9	−0.460	−0.941	−0.884	−0.057
Ganghwa-do	S2	−0.592	−1.049	−1.121	+0.073
Garorim Bay	L8	−0.720	−1.205	−1.377	+0.172
Garorim Bay	L9	−0.360	−0.824	−0.714	−0.111
Garorim Bay	S2	−0.452	−0.798	−0.884	+0.085
Gomso Bay	L8	−0.469	−0.881	−0.887	+0.006
Gomso Bay	L9	−0.406	−0.860	−0.775	−0.085
Gomso Bay	S2	−0.317	−0.637	−0.618	−0.019
Hampyeong Bay	L8	−0.313	−0.684	−0.604	−0.080
Hampyeong Bay	L9	−0.352	−0.801	−0.673	−0.128
Hampyeong Bay	S2	−0.270	−0.516	−0.528	+0.012
Suncheon Bay	L8	+0.160	+0.287	−0.020	+0.307

			Measured bias	Predicted bias	Residual
Held-out site	Sensor	$A \cdot \langle \cos \theta \rangle$ (m)	(m)	(m)	(m)
Suncheon	L9	+0.221	+0.390	+0.065	+0.325
Bay					
Suncheon	S2	+0.203	+0.312	+0.040	+0.273
Bay					

LOO summary: Pearson $r = 0.969$, RMSE = 0.156 m, MAE = 0.118 m; sign-of-bias correct in 14 / 15 held-out points (the single miss is the smallest-magnitude Suncheon L8 point; see Table S1e).

Table S2 — Robustness to amplitude definition and reference choice

Per-variant, per-sensor regression of the closed-form model **mean bias** = $\beta \cdot \mathbf{A} \cdot \langle \cos \theta \rangle$ (main-text Equation 1) under the four amplitude / reference combinations described in main-text §4.7: (a) baseline ($A = \frac{1}{2}(\text{HW} - \text{LW})$, KHOA hourly reference); (b) A = strict M_2 amplitude from a 5-year **utide** decomposition, KHOA observed reference; (c) $A = M_2$ amplitude with the reference replaced by the astronomical-only **utide.reconstruct** synthetic series; (d) FES2022b global ocean tide model used as both reference and satellite-overpass source, with A set to the site-specific mean half-range computed from the FES synthesis.

Table S2a — Pooled regression summary ($n = 15$)

Variant	β	95 % CI	Intercept	R^2	LOO	LOO	Sign
			(m)		RMSE (m)	Pearson r	correct
(a)	1.78	[1.44, 1.92]	−0.057	0.980	0.16	0.969	14/15
Base-line KHOA							
(b) M ₂	1.87	[1.50, 2.02]	−0.053	0.978	0.17	0.961	14/15
amplitude + KHOA							
(c) M ₂	1.90	[1.49, 2.06]	−0.042	0.974	0.17	0.964	14/15
amplitude + astro. ref.							
(d)	1.70	[1.42, 1.80]	+0.004	0.983	0.11	0.977	15/15
FES2022b global model							

Table S2b — Per-sensor FES2022b values (variant d)

Site	Sensor	n	A (m,			KHOA bias	FES bias
			FES)	$\langle \cos \theta \rangle$	$A \cdot \langle \cos \theta \rangle$ (m)	(m)	(m)
Ganghwa-	L8	137	1.02	−0.107	−0.109	−1.133	−0.263
do							

Site	Sensor	n	A (m,			KHOA bias	FES bias
			FES)	$\langle \cos \theta \rangle$	$A \cdot \langle \cos \theta \rangle$ (m)	(m)	(m)
Ganghwa- do	L9	90	1.02	-0.243	-0.248	-0.941	-0.419
Ganghwa- do	S2	826	1.02	-0.186	-0.190	-1.049	-0.353
Garorim Bay	L8	129	2.16	-0.143	-0.309	-1.205	-0.632
Garorim Bay	L9	91	2.16	-0.286	-0.618	-0.824	-0.987
Garorim Bay	S2	1656	2.16	-0.213	-0.461	-0.798	-0.784
Gomso Bay	L8	203	1.28	-0.233	-0.299	-0.881	-0.547
Gomso Bay	L9	110	1.28	-0.257	-0.330	-0.860	-0.562
Gomso Bay	S2	187	1.28	-0.345	-0.444	-0.637	-0.643
Hampyeong Bay	L8	195	1.48	-0.303	-0.450	-0.684	-0.700
Hampyeong Bay	L9	119	1.48	-0.251	-0.373	-0.801	-0.660
Hampyeong Bay	S2	796	1.48	-0.212	-0.314	-0.516	-0.549
Suncheon Bay	L8	70	1.00	+0.303	+0.302	+0.287	+0.523

Site	Sensor	n	A (m,			KHOA bias	FES bias
			FES)	$\langle \cos \theta \rangle$	$A \cdot \langle \cos \theta \rangle$ (m)	(m)	(m)
Suncheon	L9	38	1.00	+0.212	+0.211	+0.390	+0.410
Bay							
Suncheon	S2	435	1.00	+0.148	+0.148	+0.312	+0.317
Bay							

The FES2022b A is the mean half-range of the synthesised 5-year hourly series at each KHOA gauge coordinate, derived from the eight major constituents (M_2 , S_2 , K_1 , O_1 , N_2 , P_1 , K_2 , Q_1); the systematic under-estimate at Ganghwa-do (FES 1.02 m versus KHOA 2.83 m for the M_2 component alone) reflects the inability of the $1/30^\circ$ global grid to resolve within-bay tidal amplification; consequently the FES variant yields *lower-bound* bias magnitudes where the grid under-resolves the local geometry, while still recovering the bias sign and functional form — the basis for the gauge-free *a priori* application demonstrated in main-text §4.8.

Supplementary Figures

The four-variant robustness fit of Equation 1 of the main text (variants (a)–(d), including the FES2022b global-model variant) is summarised numerically in Table S2 above; a dedicated figure panel is omitted in this version because the per-variant β values, CIs, and LOO RMSEs are most compactly read from the table.

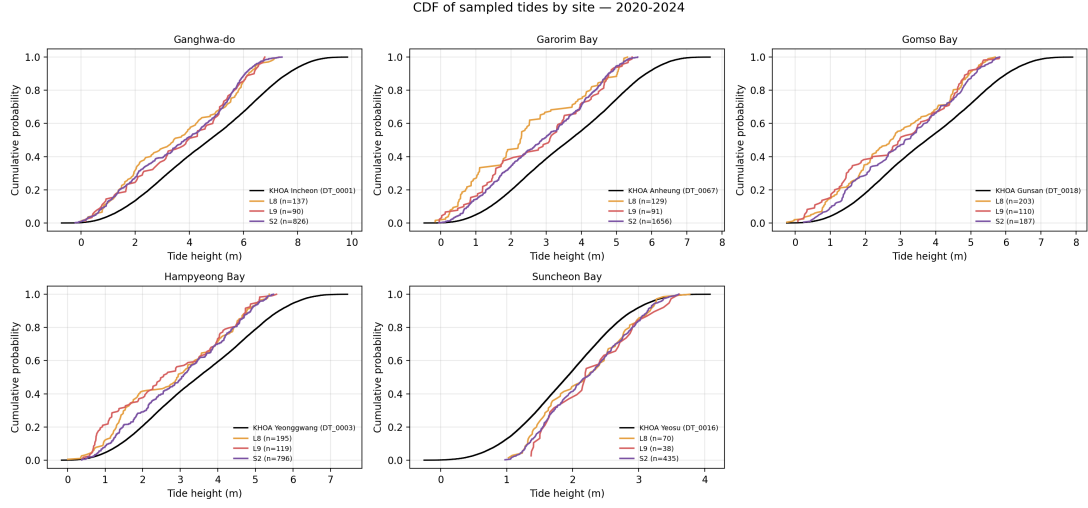


Figure S1: Cumulative distribution functions of satellite-sampled tide heights (coloured) versus the 5-year KHOA hourly reference (black) per site. Complements main-text Figure 2 (densities). The Kolmogorov–Smirnov gap between satellite and reference curves is largest at the high-tide tail on the four western sites and at the low-tide tail at Suncheon Bay.

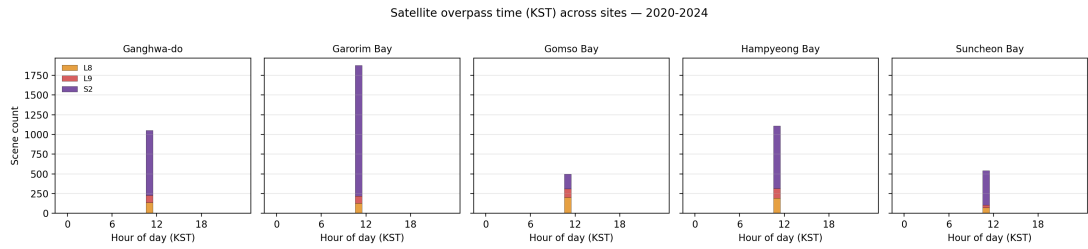


Figure S2: Hour-of-day (KST) histograms of satellite acquisition times across the five sites. Every histogram peaks at $\approx 11:00$ KST regardless of site, confirming that the spread in *tidal* phase between sites (main-text Figure 4) arises entirely from the regional amphidromic structure and not from any difference in solar overpass time.

Regression coefficient evolution — slope & R^2 stability across partitions

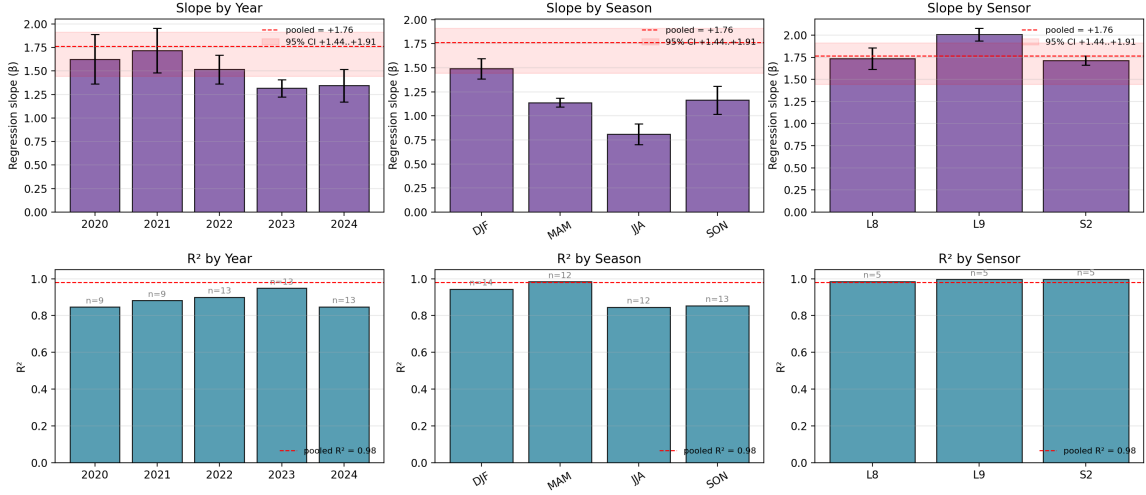


Figure S3: Stability of the regression coefficients (companion to main-text Figure 5): (top row) slope β by year / season / sensor with ± 1 SE error bars; (bottom row) R^2 by the same partitions. The horizontal red band marks the pooled ($n = 15$) bootstrap 95 % CI of β . All partitions yield β within or near the pooled CI and $R^2 \geq 0.85$, indicating that the regression is robust to year, season, and sensor and is not driven by a single subset of the data; the underlying per-(site $\times$ sensor) scatter is shown in Figure S4 and the numerical values in Table S1.

Phase $\times$ bias regression stability across years, seasons, and sensors

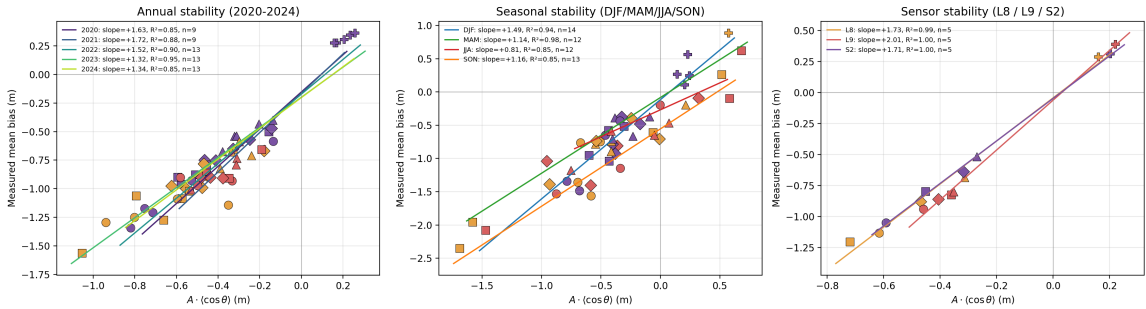


Figure S4: Scatter plots of measured mean bias versus the predictor $A \cdot \langle \cos \theta \rangle$ partitioned by year (left), meteorological season (middle), or sensor (right), each overlaid with its OLS fit. Companion to Figure S3 (which summarises the slopes and R^2 as bars).

Satellite overpass phase vs sampled tide height — 2020-2024

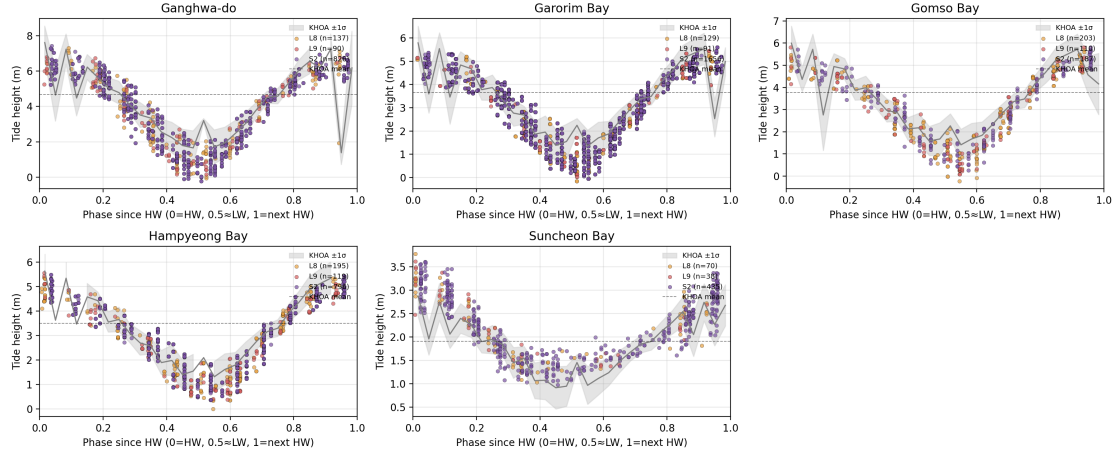


Figure S5: Per-site scatter of tide height versus normalised phase ϕ since the prior HW, with the KHOA-derived mean tide-vs-phase curve (grey band = $\pm 1\sigma$) and the overall reference mean (dashed line). Satellite acquisitions (coloured by sensor) cluster around phases that, on the western sites, lie below the reference mean (negative bias); at Suncheon Bay they lie above it (positive bias).

Quantile-mapping DEM elevation error per (site $\times$ sensor) — 2020-2024

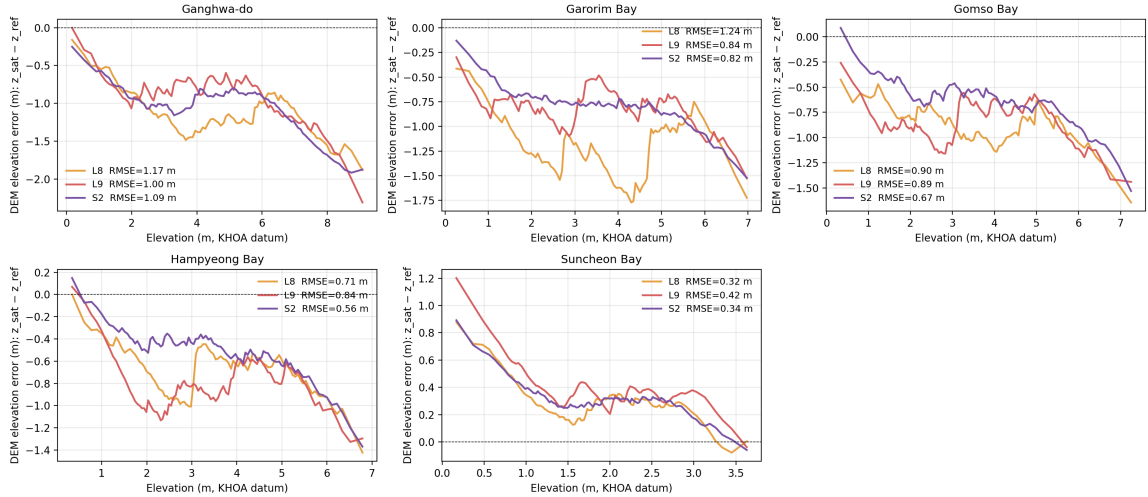


Figure S6: Elevation-domain DEM error: $z_{\text{sat}}(p) - z_{\text{ref}}(p)$ as a function of reference elevation z_{ref} , for each site (panel) and sensor (colour). The Suncheon panel is mirror-symmetric to the western panels, confirming that the tide-domain sign-reversing bias maps directly into the elevation domain of the waterline DEM. This per-quantile error underlies the per-site truncation and RMSE summaries in main-text Figure 7.

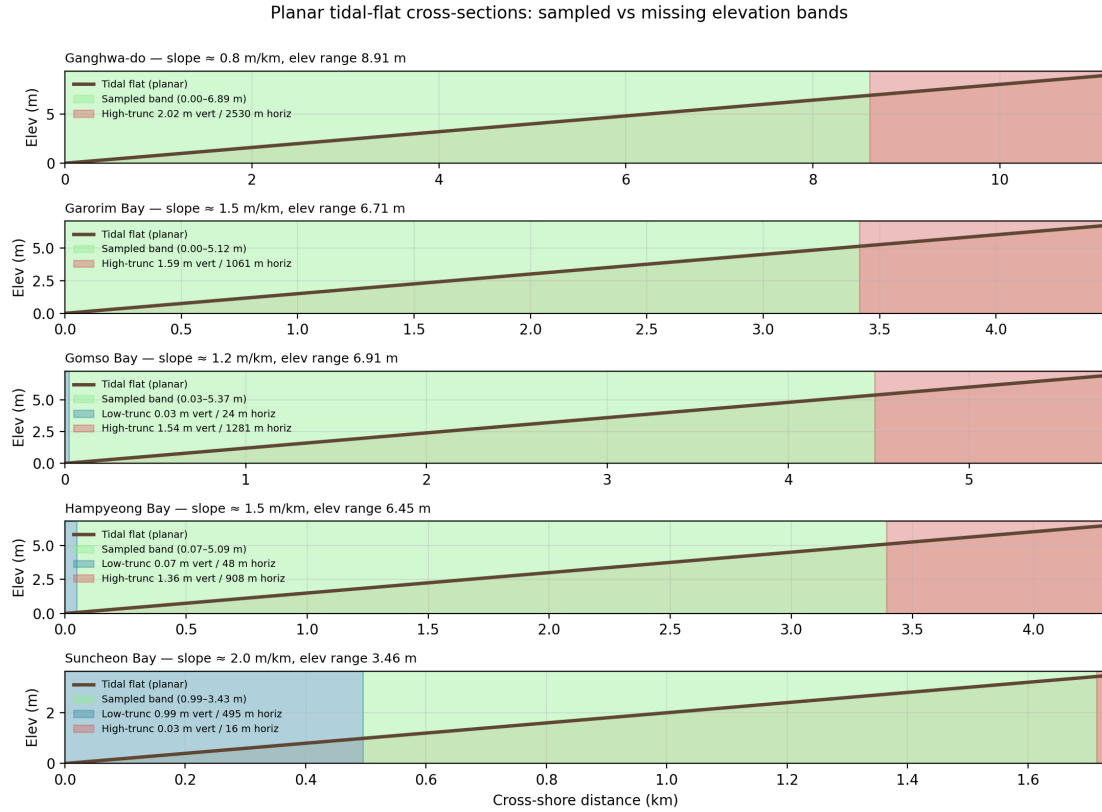


Figure S7: Schematic planar tidal-flat cross-sections per site (vertical exaggerated): brown line is the assumed planar tidal-flat surface; green shading is the elevation band sampled by satellites; red shading is the upper-tide truncation band; blue shading is the lower-tide truncation band. Annotated numbers give the vertical extent (m) and horizontal-equivalent width (m) of each band on the site-specific planar slope. Note the 2.5 km of horizontally missing upper-flat width on Ganghwa-do.